

Pearls AQI Predictor: System Architecture & Technical Documentation

Author: Mohan Singh

Track: Data Science & MLOps (10Pearls Shine Program)

Academic Institution: SZABIST University, Karachi

Live Application: <https://pearls-aqi-predictor-ai.streamlit.app/>

Repository: <https://github.com/mohansingh-ai/Pearls-AQI-Predictor>

1. Executive Summary & Problem Formulation

The Pearls AQI Predictor is an end-to-end, serverless Machine Learning Operations (MLOps) system designed to forecast ambient PM_{2.5} concentrations and derive US EPA Air Quality Index (AQI) levels for Karachi, Pakistan (24.8607° N, 67.0011° E) across direct 24-hour, 48-hour, and 72-hour forecast horizons.

Air quality in megacities is prone to severe diurnal variations and high-frequency sensor noise. To resolve this, the system incorporates a decoupled MLOps architecture that pairs scheduled feature extraction with a 7-day (168-hour) historical rolling context window, effectively eliminating single-point API anomalies and establishing a reliable environmental baseline.

2. Serverless MLOps Architecture

The infrastructure operates entirely serverless across four decoupled tiers:

- **Data Ingestion Pipeline:** Automated via GitHub Actions cron workflows, pulling hourly telemetry (temperature, relative humidity, wind speed, PM_{2.5}, PM₁₀, O₃, NO₂, SO₂, CO) from the Open-Meteo Air Quality & Weather API.
- **Hopsworks Feature Store:** Serves as the centralized feature layer (aqi_weather_features, v3), providing point-in-time consistency for retraining and online access to the 168-hour historical window.
- **Hopsworks Model Registry:** Stores and versions production multi-step Random Forest Regressor artifacts (aqi_model_day1.pkl, aqi_model_day2.pkl, aqi_model_day3.pkl).
- **Production Inference Engine:** A single-page Streamlit application hosted on Streamlit Cloud that performs real-time multi-step inference, dynamically adjusting its UI severity theme based on predicted AQI levels.

3. Key Challenges & Engineered Solutions

- **Covariate Shift & Input Volatility:** Initial single-point inputs caused severe forecast spikes during momentary sensor fluctuations. This was solved by computing the 168-hour rolling mean across all meteorological and pollutant dimensions, anchoring

models to macro environmental trends rather than micro-fluctuations.

- **48–72 Hour Slicing Truncation:** Late-night requests resulted in past hours being filtered out after array slicing, leaving the third forecast tab with only 2 hours of predictions. This was resolved by filtering elapsed timestamps prior to extracting the forward 72-hour window, ensuring a full array structure at any time of day.
- **UI Rendering Leaks:** Injecting complex nested HTML grids inside markdown containers caused raw text to leak onto the interface. This was resolved by standardizing layout cards using native Streamlit columns and moving complex analytics to an expandable container.

4. Mathematical Formulation

Rolling Window Feature Formulation:

For any feature "x" (such as temperature, humidity, wind_speed, or $PM_{2.5}$), the 168-hour rolling mean at the current reference time is computed as:

$$\mu_x(t) = (1 / 168) * \sum x(t - i) \text{ [for } i = 0 \text{ to } 167]$$

Model Input Vector:

The complete feature vector passed into the regression model is constructed as:

$$x_{\text{input}} = [\mu_{\text{temp}}(t), \mu_{\text{humidity}}(t), \mu_{\text{wind}}(t), \mu_{PM_{2.5}}(t), \text{hour}(t), \text{month}(t)]$$

EPA Piecewise Linear AQI Conversion:

Predicted $PM_{2.5}$ concentrations are mapped to the standard US EPA Air Quality Index scale via piecewise linear interpolation:

$$I = [(I_{\text{high}} - I_{\text{low}}) / (C_{\text{high}} - C_{\text{low}})] * (C - C_{\text{low}}) + I_{\text{low}}$$

5. Model Performance Benchmarks

During model selection, three candidate architectures were trained and cross-validated across all three forecast horizons. The Random Forest Regressor was selected for production due to its ability to capture non-linear meteorological interactions without overfitting.

Model Architecture	24h R^2	24h RMSE	48h R^2	48h RMSE	72h R^2	72h RMSE
Random Forest (Production)	0.92	0.85	0.84	1.25	0.78	1.80
XGBoost Regressor	0.89	0.91	0.81	1.34	0.74	1.95

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Ridge Regression (Baseline)	0.65	1.45	0.52	1.87	0.46	2.31

6. SHAP Feature Attribution

Global feature importance was computed using TreeSHAP values to determine the primary drivers of model output:

- **PM_{2.5} Rolling Mean (8.77):** Serves as the primary anchor, preventing seasonal drift.
- **Current AQI Baseline (5.52):** Provides short-term momentum for the initial 24-hour horizon.
- **Month & Meteorology (1.46 - 1.75):** Dictates boundary conditions, such as low winter wind speeds preventing pollutant dispersion versus high summer marine winds clearing particulate matter.